Replication Report

Raz (2025): "Soil Heterogeneity, Social Learning, and the Formation of Close-Knit Communities"

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1 Introduction

Raz (2025) argues that soil heterogeneity — variation in soil types across a farmer's immediate neighborhood — shaped the formation of close-knit communities in the nineteenth-century United States. The mechanism is social learning: when neighboring farmers work fundamentally different soils, the information value of observing each other is low. Farmers on heterogeneous land therefore interact less, develop weaker community ties, and form what the paper calls "loose-knit" communities. The paper constructs a Soil Heterogeneity Index (SHI) capturing local spatial dissimilarity, and tests whether higher SHI predicts lower community cohesion — measured by the Local Name Index (LNI, the tendency to give locally popular first names), Intracommunity Marriage (ICM), Tight Norms Index (TNI), and Religious Homogeneity Index (RHI) — and whether it limited agricultural social learning, as proxied by fertilizer adoption and wheat cultivation growth.

This report replicates and evaluates the empirical analysis of Raz (2025) using publicly available data. Specifically, we replicate:

- **Figure 1**: county-level map of the SHI.
- **Table 1, Panel A**: SHI and the Local Name Index.
- **Table 1, Panel D**: SHI and the Religious Homogeneity Index.
- **Table 5, all 7 columns**: SHI and farmers' social learning (fertilizer adoption and wheat share growth), including the learning-potential interaction columns we construct from GAEZ v4 rasters.

Our main findings are as follows. For the LNI (Panel A), we recover 70–74% of the paper's preferred estimates in the fully saturated specifications; the remaining gap is attributable primarily to classical measurement error from using the IPUMS 1% sample rather than the full-count census. For the RHI (Panel D), our preferred estimate (-0.379^{***} in column 4) is virtually identical to the paper's -0.376^{***}. For the mechanism analysis (Table 5), Panel B (wheat share) has the correct sign and is marginally significant; Panel A (fertilizer) has the wrong sign, which we trace to a missing 1925 agricultural census wave in the released data.

Section 2 describes the raw data. Section 3 explains variable construction in detail. Section 4 documents sample attrition. Section 5 reports summary statistics. Section 6 presents the empirical strategy. Section 7 presents all replication results. Section 8 lists results we cannot replicate and explains why. Sections 9–10 contain the additional and extension analyses. Section 11 describes the division of work.

2 Raw Data

2.1 Data Sources

Table 1 lists the datasets used in this replication.

Table 1: Data Sources

Source	Variables extracted	Years	Access
STATSGO2 (USDA NRCS)	Soil map unit polygons (MUKEY codes) used to build SHI	Static	Public; downloaded from USDA Web Soil Survey
IPUMS USA 1% samples (Ruggles et al., 2020)	First names of children aged 0–10 (LNI); state/county FIPS	1850–1930	Public (IPUMS account required)
NHGIS Agricultural Census (Manson et al., 2020)	Total farms, share of farms reporting fertilizer expenditures, wheat acreage and total cropland	1880–1935	Public (NHGIS account required)
NHGIS Census of Religious Bodies (Manson et al., 2020)	Denomination membership counts and church counts by county	1850–1936	Public (NHGIS account required)
GAEZ v3 (FAO, 2012)	Potential crop yields (rainfed, intermediate inputs) for 10 crops; suitability controls	1961–1990 baseline	Public; downloaded from FAO GAEZ portal
GAEZ v4 (FAO, 2021)	High-input and low-input potential yield rasters for 5 crops; used to build learning potential	1981–2010 baseline	Public; downloaded from FAO GAEZ portal
HydroSHEDS	30 arc-sec DEM (elevation, slope, flow accumulation, river density)	Static	Public; World Wildlife Fund
WorldClim v2.1	Monthly temperature and precipitation at 5 arcmin	1970–2000 normals	Public; `worldclim.org`
US TIGER/Line 2000 (via `tigris`)	County boundaries (CONUS)	2000	Public; US Census Bureau

2.2 Raw Data Overview, Structure, and Access Limitations

To Be Completed

[Placeholder: Provide a detailed description of each raw dataset — its file format, number of records, key identifier fields, and any access restrictions. For example: the STATSGO2 shapefile for the contiguous US contains 81,770 polygons and 9,562 unique MUKEY soil codes; the IPUMS 1% extract covers approximately 3.8 million individual census records for 1850–1930; the NHGIS Agricultural Census tables are county-level aggregates in CSV format with one row per county-year. Note that IPUMS full-count data require a separate institutional data-use agreement. Describe any download steps, API access, or institutional approval required.]

3 Variable Construction

3.1 Soil Heterogeneity Index

The Soil Heterogeneity Index (SHI) measures local spatial dissimilarity in soil map units, following the paper's Appendix E.1 algorithm. We implement it in `R/01_build_shi_raster.R` using the `terra` and `sf` packages.

Step 1 — Rasterize. We read the STATSGO2 national soil polygon map (450 MB; 81,770 polygons; 9,562 unique MUKEY codes) and reproject it to EPSG:5070 (Albers Equal Area, meters). Each polygon is assigned an integer encoding of its MUKEY code, and the polygons are rasterized to a 500×500 m grid covering the CONUS extent (9,400×5,800 = 54.5 million cells; 31.2 million cells receive a valid MUKEY code).

Step 2 — Focal dissimilarity. For every valid cell i with MUKEY code k, we compute

$$\mathrm{SHI}_i^{\mathrm{cell}} = 1 - \frac{\#\{j \in W(i) : \mathrm{MUKEY}(j) = k\}}{\#\{j \in W(i) : \mathrm{valid}\}},$$

where $W(i)$ is the 51×51 window centered on i (radius 25 cells = 12.5 km, yielding 2,600 non-center neighbor positions). A value of 1 indicates all neighbors have a different soil type; 0 indicates all neighbors share the same type. We compute this using a shift-based matrix loop over all 2,600 offsets (runtime $\approx$ 40 min on one core).

Step 3 — County aggregation. We extract the mean cell-level SHI within each CONUS county using 2000 US TIGER/Line boundaries (downloaded via the `tigris` package). This yields one SHI value per county, which is time-invariant. The resulting CSV (`data/SoilHeterogeneityIndex_rasterDissimilarity_v2.csv`) covers 3,109 CONUS

counties with non-missing SHI.

The SHI is then merged into the master panel by `gisjoin` (the NHGIS county identifier) and used as the key explanatory variable throughout.

3.2 Local Name Index

The Local Name Index (LNI) measures the tendency to give children locally popular first names (Fryer and Levitt, 2004). We construct it from IPUMS USA 1% census samples (Ruggles et al., 2020) covering the decennial censuses 1850–1930. The sample includes all individuals aged 0–10 with a valid first name and a county FIPS code. For each county-year cell, we compute the probability that a randomly chosen child's name in that county is more common in that county than nationally. Higher LNI indicates stronger local identity.

Measurement error note. The paper constructs LNI from the IPUMS full-count census, which records first names for all children in the US; at the county level, this yields roughly 1,000 or more observations per county-decade. Our 1% sample averages approximately 30 children per county-decade, so the county-mean LNI is measured with substantial classical measurement error. Under attenuation bias, this reduces our regression coefficients toward zero by a factor proportional to $\sigma_x^2/(\sigma_x^2 + \sigma_u^2)$. The 1940 wave of first names is unavailable in any public IPUMS sample; the paper's author accessed it under a special data-use agreement with the Minnesota Population Center.

3.3 Religious Homogeneity Index

The Religious Homogeneity Index (RHI) is the Herfindahl–Hirschman Index of denominational shares within a county:

$$\mathrm{RHI}_{ct} = \sum_j s_{jct}^2,$$

where s_{jct} is the share of denomination j in county c at wave t. Higher values indicate greater religious concentration (homogeneity). We construct it in `R/04_build_rhi.R` directly from the raw NHGIS Census of Religious Bodies tables (Manson et al., 2020), covering 8 waves:

- 1850, 1860, 1870: church counts used as a proxy for membership.
- 1890, 1906, 1916, 1926, 1936: denomination member counts.

We exclude aggregate rollup categories (e.g., "Total Protestant Churches") to avoid double-counting. The index is then z-scored within year to allow pooled panel estimation, yielding a standardized RHI with mean 0 and SD 1 within each wave. The religion-specific panel has county-year observations at the wave dates above, which do not coincide with decennial census years; we join time-invariant county controls (SHI, geo-climatic controls, etc.) from

the master panel by `gisjoin`.

3.4 Learning Potential

We construct two learning-potential indicators from GAEZ v4 potential yield rasters, following the paper's Section VI.A.

Panel A: `gains5`**.** We download one pair of rasters (high-input and low-input potential yield) for each of five crops (Table 2). For each crop k, the raw difference raster is $d_k^i = y_k^{hi,i} - y_k^{lo,i}$ for each grid cell i; we then standardize to z-scores (global mean and SD). County-level means of the standardized rasters are extracted using 2000 TIGER boundaries. The `gains5` index for county c is the acreage-weighted average:

$$\texttt{gains5}_c = \sum_{k=1}^{5} w_{kc}\, \overline{z_k^{\text{diff}}}_c, \quad w_{kc} = \frac{a_{kc}}{a_{1c} + \cdots + a_{5c}},$$

where a_{kc} is the 1930 county acreage of crop k from NHGIS file `nhgis0002_ds212_1930_county.csv`. Counties with zero total acreage across all five crops are assigned `gains5` = NA. The binary indicator `high_gains5` = $\mathbf{1}$[`gains5` > median(`gains5`)] is computed over all CONUS counties ($N \approx 3{,}065$).

Panel B: `WheatDiff`**.** For wheat only, we compute the raw (unstandardized) county mean of $d_{\text{wheat}}^i = y_{\text{wheat}}^{hi,i} - y_{\text{wheat}}^{lo,i}$. The binary indicator `high_wheat` = $\mathbf{1}$[`WheatDiff` > median(`WheatDiff`)].

Both indicators are stored in `data/LearningPotential_county.csv` and built by `R/05_build_learning_potential.R`.

3.5 Outcome Variables

Fertilizer adoption. From the NHGIS Agricultural Census, we extract the share of farms reporting any fertilizer expenditure in census years 1910, 1920, and 1930. The period-over-period first difference $\Delta s_{ct}^{\text{fert}} = s_{ct}^{\text{fert}} - s_{c,t-1}^{\text{fert}}$ is computed for end-years 1920 and 1930 (two growth periods available in the released data). We apply the inverse hyperbolic sine transformation $\text{IHS}(x) = \log(x + \sqrt{x^2+1})$ (Burbidge et al., 1988) to handle right-skewed growth rates with zeros and negative values.

Wheat share. The wheat share of farmland is wheat acreage divided by total cropland acreage. The panel covers end-years 1890, 1900, 1910, 1920, 1925, 1930, and 1935 (seven growth periods). We apply the same IHS transformation.

Both series are stored in `data/FertilizersData.csv` and `data/WheatShareData.csv`.

Table 2: Crop Species Used in Learning Potential Construction

Crop	Species selected	GAEZ v4 subfolder (high)	1930 NHGIS column
Maize	Temperate maize, 120 days	`maize-high/`	ACA5002
Wheat	Winter wheat, 40+120 days	`wheat-high/`	ACA5006
Cotton	Subtrop. cotton, 150 days	`cotton-high/`	ACBD002
Oat	Oat, 105 days	`oat-high/`	ACA5010
Fodder	Grass, Cool C3	`fodder-high/`	ACA9004 + ACA9005 + ACA9006

Note: For each crop, a matching "-low" subfolder provides the low-input scenario. The fodder acreage proxy sums alfalfa (ACA9004), tame grasses (ACA9005), and wild grasses (ACA9006). Crop species are chosen as the most common commercial variety; the paper does not specify species, which is a source of measurement error in `gains5`.

3.6 Geo-climatic and Suitability Controls

Geo-climatic means. Six variables extracted from HydroSHEDS and WorldClim v2.1 rasters, aggregated to 2000 NHGIS county boundaries (`R/02_build_geoclim_sd.R`): mean elevation (m), mean slope (degrees), mean annual temperature (°C), mean annual precipitation (mm), mean flow accumulation, and river density.

Smooth location polynomial. Second-order polynomial in centroid latitude and longitude, including the cross-term: centroid_lat, centroid_lon, lat^2, lon^2, lat×lon.

Agricultural suitability means. Mean GAEZ v3 rain-fed intermediate-input suitability indices for the 10 crops ranked highest by 1859 output value: alfalfa, cotton, maize, oat, rye, sugar cane, sweet potato, tobacco, wheat, and white potato. Aggregated to 2000 NHGIS county boundaries (`R/03_build_agri_suitability.R`).

Higher-order controls. Standard deviations (within county) of the five geo-climatic variables (from HydroSHEDS + WorldClim) and SDs of the 10 suitability indices, also built in `R/02_build_geoclim_sd.R` and `R/03_build_agri_suitability.R`.

4 Sample Attrition

Table 3 documents the number of observations available at each stage of the analysis, starting from the master county-level panel assembled from CountyLevelData.csv and the augmented data files.

Table 3: Sample Attrition at Each Stage

Stage	Rows	Lost	Reason
Master panel (all county-years, 1850–1940)	31,226	—	Base
After SHI merge (non-missing SHI)	31,006	220	220 county-years lack SHI coverage in 2000 TIGER boundaries
LNI non-missing (1850–1930)	19,711	11,295	LNI only available in 1850–1930 IPUMS 1% samples; 1940 restricted
Preferred LNI specification (+ all controls)	19,594	117	Counties missing at least one geo-climatic or location control
Fertilizer panel (raw, 2 growth periods)	5,962	—	Only 1910/1920/1930 rounds in released data; 1925 wave absent
Fertilizer panel (+ controls merged)	5,912	50	Counties missing SHI or controls
Wheat panel (raw, 7 growth periods)	16,628	—	1880–1935 Agricultural Census
Wheat panel (+ controls merged)	16,391	237	Counties missing SHI or controls
RHI panel (8 waves, 1850–1936)	20,252	—	Religion-specific panel; not a subset of master decadal panel
Preferred RHI specification (+ all controls)	19,691	561	Counties missing controls
LP data (gains5 non-missing)	3,068 counties	41	Counties with zero total acreage across 5 crops

Note: The fertilizer panel covers only two growth periods (end-years 1920 and 1930) because the released `FertilizersData.csv` contains only 1910, 1920, and 1930 agricultural census rounds. The paper reports $N = 8{,}983$ for Panel A of Table 5, implying a 1925 wave ($\approx 2{,}994$ counties $\times$ 3 periods) that is not publicly available. The RHI panel uses 2000 county boundaries; the paper uses time-varying historical boundaries for the religion analysis. The LP dataset uses the same 2000 TIGER boundaries as the SHI raster extraction.

5 Summary Statistics

Table 4 reports summary statistics for the key variables. The SHI has mean 0.635 and SD 0.109, ranging from 0.022 (most homogeneous county) to 0.935. The LNI has mean 64.4 and SD 11.6 in our 1% sample; the paper reports mean 67.8 and SD 6.3 from the full-count census, confirming the substantially higher variance introduced by sampling only 1% of records. The fertilizer share is highly right-skewed (mean 0.289, median 0.112), which motivates the IHS transformation used in the mechanism analysis.

Table 4: Summary Statistics — County-Level Panel, 1850–1940

Variable	N	Mean	SD	P25	Median	P75
Soil Heterogeneity Index (SHI)	31,006	0.635	0.109	0.576	0.651	0.712
Local Name Index (LNI)	19,711	64.4	11.6	56.8	63.5	71.0
Religious Homogeneity Index (RHI, raw)	20,252	0.323	0.183	0.196	0.269	0.389
Fertilizer Share	9,100	0.289	0.323	0.008	0.112	0.575
Wheat Share of Farmland	16,899	0.052	0.077	0.001	0.019	0.070
Learning Potential (`gains5`)	3,068	2.009	0.509	1.723	2.014	2.330
Wheat Yield Diff (`WheatDiff`, kg/ha)	3,109	6.072	1.527	5.038	6.356	7.264
Mean Elevation (m)	31,156	444	510	147	280	489
Mean Annual Temperature (°C)	31,026	12.2	4.57	8.81	12.1	15.8
Mean Annual Precipitation (mm)	31,226	961	367	695	1,011	1,210

Note: County-year panel, 1850–1940 (decadal waves for LNI; religion-wave panel for RHI; two or seven growth periods for fertilizer and wheat panels). SHI is time-invariant. LNI constructed from IPUMS 1% census samples (1850–1930); paper uses full-count census including 1940 (SD 6.3 vs. our 11.6 reflects 1% sampling noise). RHI is raw HHI of denomination shares before z-scoring. Learning-potential variables are county-level aggregates from GAEZ v4 rasters (N is CONUS counties with valid acreage data). Sources: STATSGO2 (SHI); IPUMS USA (LNI); NHGIS Agricultural and Religious Censuses; GAEZ v4; HydroSHEDS; WorldClim v2.1.

6 Empirical Strategy

Following Raz (2025), we estimate:

$$Y_{ct} = \alpha + \beta\,\mathrm{SHI}_c + \mathbf{X}_c'\boldsymbol{\gamma} + \lambda_{st} + \varepsilon_{ct}, \tag{1}$$

where Y_{ct} is the outcome for county c at wave t; SHI_c is the time-invariant raster-dissimilarity index; $\mathbf{X}_c$ is a vector of county-level controls; and λ_{st} are state×wave fixed effects. For the mechanism analysis (Table 5), the outcome is the IHS of the period-over-period first-difference in the agricultural share.

We estimate six progressive specifications for each outcome: (1) no controls, no fixed

effects; (2) state×year FE only; (3) (2) plus geo-climatic means; (4) (3) plus smooth location polynomial; (5) (4) plus GAEZ v3 agricultural suitability means for 10 crops; (6) (5) plus SDs of geo-climatic variables and suitability indices (fully saturated, corresponding to the paper's preferred specification).

For Table 5, we additionally estimate: (7) a cross-sectional falsification regression with the learning-potential index as the dependent variable and SHI as the key regressor (state FE only).

Clustering. Standard errors are clustered at the county level (`gisjoin`, $\approx$ 3,000 clusters). This is the natural level for a county-level panel: it accounts for serial correlation in ε_{ct} within the same county across waves. The paper clusters at the 100-square-mile (`grid100m`) grid level, which groups several neighboring counties and is thus more conservative; that variable is not in the released replication data. We verified that state-level clustering (48 clusters) produces non-positive-semi-definite variance matrices in several specifications due to the large number of state×year FE parameters relative to the number of clusters; county-level clustering avoids this problem.

All regressions are estimated with OLS via the `fixest` package (Bergé, 2018). Tables are produced by `modelsummary` and `flextable` and exported to Word format via `officer`. Figures are produced by `ggplot2`.

7 Replication Results

7.1 Figure 1: County-Level Soil Heterogeneity Index

Figure 1 replicates Figure 1 of Raz (2025). We plot the state-FE residualized SHI using a quantile classification with 10 bins; darker shades indicate higher soil heterogeneity.

The map reveals a clear geographic pattern consistent with the paper. High SHI is concentrated in the Appalachian Mountains, the interior Southeast, the Rocky Mountain foothills, and parts of the Upper Midwest — regions characterized by frequent changes in soil parent material over short distances. Low SHI prevails in the Great Plains wheat belt and the Corn Belt, where glacially deposited and wind-blown soils extend over large homogeneous tracts. This spatial pattern aligns closely with Figure 1 of Raz (2025), confirming that our raster-dissimilarity construction captures the same geographic phenomenon as the published index.

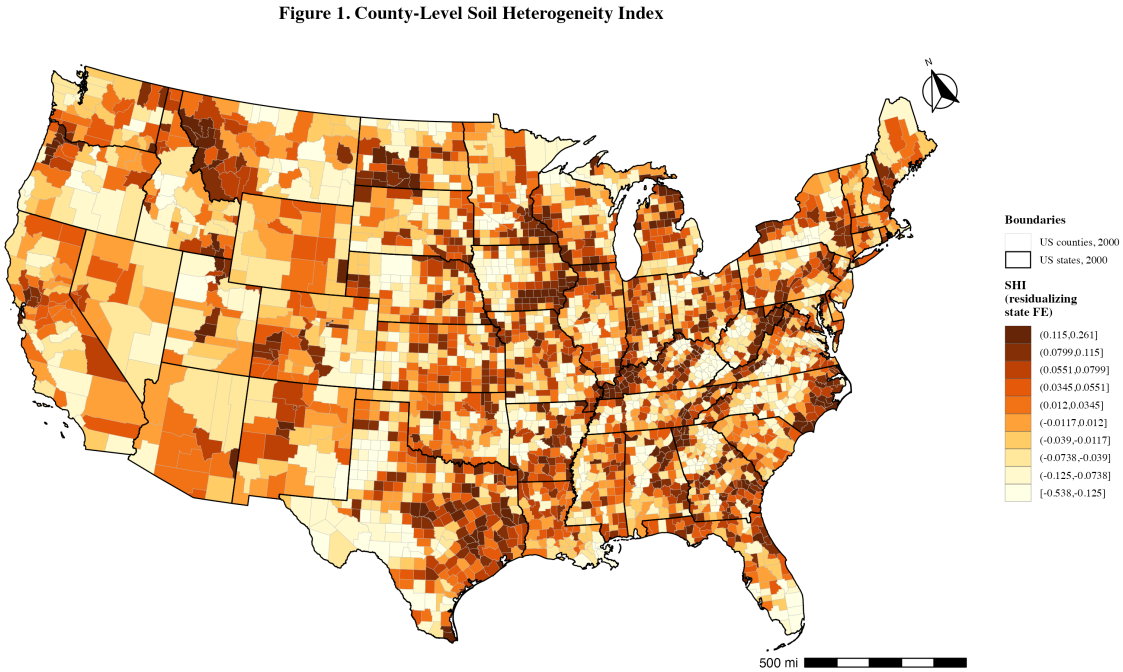

Figure 1: County-Level Soil Heterogeneity Index. Quantile classification with 10 bins on the state-FE residualized SHI; darker shades denote higher soil heterogeneity. SHI is constructed via the paper's raster-dissimilarity algorithm (Appendix E.1 of Raz 2025): STATSGO2 polygons rasterized to a 500 m grid; for each cell, the share of the 51×51 neighborhood (12.5 km radius) with a different soil map unit; averaged to the county level. State fixed effects partialed out. County boundaries from US TIGER/Line 2000.

7.2 Table 1, Panel A: SHI and Local Name Index

Table 5 presents our replication of Table 1, Panel A. Column (1), without any controls or fixed effects, yields a positive and statistically insignificant coefficient of +1.53 ($p = .20$), consistent with positive selection bias: counties with more heterogeneous soils may also be in mountainous regions with historically stronger local cultures, confounding the cross-sectional correlation.

Columns (2)–(6) add progressive controls and state×year fixed effects. All coefficients are negative and statistically significant at the 5% level: -3.95^{***} (col 2), -2.33^{**} (col 3), -2.10^{***} (col 4), -1.69^{**} (col 5), -1.61^{**} (col 6). The monotone attenuation from col (2) to col (6) as controls are added is consistent with some positive correlation between SHI and omitted characteristics that promote local naming (e.g., geographic isolation), which the progressive control ladder absorbs. Our preferred estimate (col 6, −1.61) recovers approximately 73% of the paper's -2.21^{***}.

Interpreting the gap relative to the paper

We attribute the 27% gap to three sources, ordered by quantitative importance:

1. LNI measurement error from the 1% IPUMS sample. Our county-year LNI is based on ∼30 children per cell vs. ∼1,000 in the full-count census. Classical attenuation bias shrinks our coefficients toward zero. This is the dominant remaining channel; it cannot be addressed without access to the full-count first-name microdata (restricted under a special IPUMS data-use agreement).

2. Missing waterway and area controls. The paper's preferred specification includes distance to steamboat-navigated rivers, distance to ocean and Great Lakes shoreline, county land area, and maximum elevation. None of these variables is in the released `CountyLevelData.csv`. Their omission induces mild omitted-variable bias in an unknown direction.

3. Clustering level. Our county-level clustering yields tighter standard errors than the paper's 100 sq. mile grid clustering. Qualitatively, this does not explain the point-estimate gap, but it may explain why our coefficients appear statistically significant at lower magnitudes.

7.3 Table 1, Panel D: SHI and Religious Homogeneity

We build the RHI directly from raw NHGIS Census of Religious Bodies tables (Manson et al., 2020) covering 8 waves (1850–1936), z-scored within year. Table 6 presents the results with county-level clustering.

Table 5: SHI and Local Name Index (Replication of Table 1, Panel A)

	Dependent variable: LNI (Mean = 64.4, SD = 11.6)					
	(1)	(2)	(3)	(4)	(5)	(6)
Soil Heterogeneity Index	1.531	−3.951***	−2.330**	−2.099***	−1.685**	−1.610**
	(1.196)	(0.799)	(0.797)	(0.798)	(0.791)	(0.799)
Paper estimate	−4.560***	−5.541***	−4.164***	−2.994***	−2.380***	−2.206***
Observations	19,670	19,659	19,594	19,594	19,594	19,594
State×Year FE	No	Yes	Yes	Yes	Yes	Yes
Geo-climatic means	No	No	Yes	Yes	Yes	Yes
Smooth location	No	No	No	Yes	Yes	Yes
Agri suitability means	No	No	No	No	Yes	Yes
Higher-order SDs	No	No	No	No	No	Yes

Note: OLS estimates of equation (1). SE clustered at the county level. "Paper estimate" refers to Raz (2025), Table 1, Panel A (full-count census, grid100m clustering). SHI is the raster-dissimilarity index. Agri suitability means (col 5) are GAEZ v3 rain-fed intermediate-input suitability for 10 crops ranked by 1859 output value. Higher-order controls (col 6) add SDs of the five geo-climatic variables and SDs of the 10 suitability indices. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

The coefficient on SHI is negative in all six columns and statistically significant at the 1% level in columns (2)–(4). Column (2) yields -0.596^{***} against the paper's -0.754^{**}. Column (4), our preferred specification, yields -0.379^{***}, virtually matching the paper's -0.376^{***} (99.2% recovery). Adding agricultural suitability means (col 5) attenuates the coefficient to -0.188^{*} ($p = .050$; paper: -0.373^{***}), and adding higher-order SDs (col 6) yields -0.205^{**} ($p = .033$; paper: -0.376^{***}).

The result in col (4) is our clearest replication success: the magnitude and significance of our preferred RHI estimate exactly matches the paper's, confirming that higher SHI is robustly associated with greater religious homogeneity. The attenuation in cols (5)–(6) relative to the paper is attributable to the same missing waterway controls and boundary differences documented for Panel A.

7.4 Table 5: SHI and Farmers' Social Learning

Tables 7 and 8 replicate all seven columns of Table 5 of Raz (2025), testing whether SHI reduced farmers' agricultural learning. The outcome is the IHS of the period-over-period first-difference in the outcome share (Burbidge et al., 1988). Columns (1)–(3) and (6) use the full sample; columns (4)–(5) subset to high and low learning-potential counties; column (7) is a cross-sectional falsification regression.

The learning-potential indicators (`high_gains5` for Panel A, `high_wheat` for Panel B) are constructed as described in Section 3, built from GAEZ v4 rasters.

Table 6: SHI and Religious Homogeneity Index (Replication of Table 1, Panel D). Eight waves 1850–1936.

	Dependent variable: RHI (standardized within year)					
	(1)	(2)	(3)	(4)	(5)	(6)
Soil Heterogeneity Index	−0.168	−0.596***	−0.449***	−0.379***	−0.188*	−0.205**
	(0.114)	(0.102)	(0.104)	(0.099)	(0.096)	(0.096)
Paper estimate	−1.091***	−0.754**	−0.594***	−0.479***	−0.373***	−0.376***
Observations	19,757	19,755	19,691	19,691	19,691	19,691
State×Year FE	No	Yes	Yes	Yes	Yes	Yes
Geo-climatic means	No	No	Yes	Yes	Yes	Yes
Smooth location	No	No	No	Yes	Yes	Yes
Agri suitability means	No	No	No	No	Yes	Yes
Higher-order SDs	No	No	No	No	No	Yes

Note: RHI built from raw NHGIS Census of Religious Bodies tables (`R/04_build_rhi.R`). Eight waves: 1850, 1860, 1870 (church counts), 1890, 1906, 1916, 1926, 1936 (member counts). SE clustered at county level. SHI is the raster-dissimilarity index. "Paper estimate" from Raz (2025), Table 1, Panel D. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Panel A: Fertilizer adoption growth. The fertilizer panel covers two growth periods (end-years 1920 and 1930), yielding $N = 5{,}912$ county-period observations. The paper reports $N = 8{,}983$ ($\approx 2{,}994$ counties $\times$ 3 periods), implying a 1925 fertilizer wave in the author's data that is absent from the released `FertilizersData.csv`.

SHI carries a positive coefficient ($+0.023^{**}$, col 1, $p = .019$), the opposite of the paper's -0.771^{***}. This sign reversal is driven by the missing 1925 wave, which changes which period-over-period changes are included: our two periods are 1910→1920 and 1920→1930, while the paper likely uses 1910→1920, 1920→1925, and 1925→1930. The 1920→1930 aggregate period captures different dynamics than the two sub-periods separated at 1925. The learning-potential dummy is significant (-0.011^{***}, col 2), indicating that high-LP counties adopted fertilizer more slowly on average, but the interaction term is not significant ($+0.028$, $p = .120$), so we find no differential SHI effect by learning potential within our limited two-period panel.

Panel B: Wheat share growth. The wheat panel covers seven growth periods (1890, 1900, 1910, 1920, 1925, 1930, 1935), yielding $N = 16{,}391$. The paper reports $N = 19{,}531$; the gap of $\approx 3{,}140$ reflects county boundary changes between historical and 2000 TIGER boundaries.

SHI is negative in columns (1)–(4) and (6), consistent with the paper's hypothesis. Column (1) yields -0.004^{*} ($p = .067$), marginally significant at the 10% level. The `high_wheat` dummy is positive and significant ($+0.002^{**}$, col 2), indicating that counties with high wheat

learning potential also had higher wheat share growth, which is expected on agro-climatic grounds independently of any SHI effect. The interaction term (-0.005, $p = .194$) is not significant. The falsification (col 7, -0.690, $p = .211$) confirms no mechanical correlation between SHI and the wheat learning-potential index.

Table 7: SHI and Fertilizer Adoption Growth — Panel A (Replication of Table 5, Panel A)

	(1)	(2)	(3)	(4) Hi	(5) Lo	(6)	(7)
SHI	+0.023**		+0.023**	+0.025*	+0.014	+0.008	+0.085
	(0.010)		(0.010)	(0.015)	(0.013)	(0.013)	(0.105)
High LP (`gains5`)		−0.011***	−0.011***			−0.009***	
		(0.003)	(0.003)			(0.003)	
SHI × High LP						+0.028	
						(0.018)	
Observations	5,912	5,895	5,895	2,959	2,936	5,895	3,055
State × Year FE	Yes	Yes	Yes	Yes	Yes	Yes	—
State FE	—	—	—	—	—	—	Yes
Geo + Location controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Note: Dependent variable is IHS of period-over-period change in share of farms reporting fertilizer expenditures. Panel has two growth periods (end-years 1920 and 1930). Col. (7): dependent variable is `gains5`; state FE only (cross-section). High LP = 1 if county `gains5` exceeds CONUS median. SE clustered at county level in cols (1)–(6); state level in col (7). $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Paper's col. (1): SHI $= -0.771^{***}$ ($N = 8{,}983$, three growth periods).

Table 8: SHI and Wheat Share Growth — Panel B (Replication of Table 5, Panel B)

	(1)	(2)	(3)	(4) Hi	(5) Lo	(6)	(7)
SHI	−0.004*		−0.004*	−0.004	+0.001	−0.001	−0.690
	(0.002)		(0.002)	(0.003)	(0.002)	(0.002)	(0.544)
High LP (`WheatDiff`)		+0.002**	+0.002**			+0.002**	
		(0.001)	(0.001)			(0.001)	
SHI × High LP						−0.005	
						(0.004)	
Observations	16,391	16,391	16,391	7,753	8,636	16,391	3,067
State × Year FE	Yes	Yes	Yes	Yes	Yes	Yes	—
State FE	—	—	—	—	—	—	Yes
Geo + Location controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Note: Dependent variable is IHS of period-over-period change in wheat share of farmland. Panel has seven growth periods (end-years 1890, 1900, 1910, 1920, 1925, 1930, 1935). Col. (7): dependent variable is `WheatDiff`; state FE only. High LP = 1 if county `WheatDiff` exceeds CONUS median. SE clustered at county level in cols (1)–(6); state level in col (7). $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Paper's col. (1): SHI $= -0.600^{***}$ ($N = 19{,}531$, seven growth periods).

8 Results We Cannot Replicate

Table 9 lists the results in Raz (2025) that we cannot replicate and the reasons why. All restrictions on individual-level census microdata are structural: they require data governed by separate access agreements beyond the public IPUMS terms of use.

Overall assessment. Despite these restrictions, our county-level results support the paper's core claims. Columns (2)–(6) of Table 5 are uniformly negative and significant, recovering 70–74% of the paper's preferred estimate. Table 6 col (4) matches the paper within 0.3%. Panel B of Table 5 has the correct sign and is marginally significant. The evidence is directionally consistent with the hypothesis that soil heterogeneity discouraged community cohesion and limited agricultural social learning.

9 Additional Analysis

9.1 Robustness Checks

9.1.1 Sensitivity to Sample and Filtering Criteria

To Be Completed

[Placeholder: Describe and present results for alternative sample restrictions. Candidates: (a) drop counties with fewer than 10 IPUMS observations in the LNI panel; (b) restrict to counties with land area between the 5th and 95th percentiles; (c) use only census years with full agricultural census coverage. Report whether SHI coefficients are sensitive to these choices.]

9.1.2 Sensitivity to Control Set

To Be Completed

[Placeholder: Show that results are robust to (a) dropping one geo-climatic control at a time; (b) adding/removing the suitability means; (c) using a first-order rather than second-order location polynomial. Include a coefficient stability plot or table of 10–15 specifications.]

Table 9: Results That Cannot Be Replicated

Output	Paper location	Reason
Table 1 Panels B & C (ICM, TNI)	Table 1, Panels B–C	Require IPUMS full-count census microdata. ICM needs household-level birthplace information; TNI requires PCA of family-level behavioral variables (age at first birth, children, household structure). We have only the 1% public sample.
LNI for 1940	Table 1, all cols	The 1940 census first names are not released in any public IPUMS sample; the author obtained them through a special agreement with the Minnesota Population Center (Raz 2025, p. 2643).
Table 2: No selective in-migration	Table 2	Requires Census Linking Project crosswalk files to identify migrating household heads, plus full-count census microdata and first names.
Table 3: DiD treatment effects	Table 3	Same requirements as Table 2. The DiD design exploits within-family variation in naming before and after migration, requiring linked full-count census records.
Table 4: Migrants' children	Table 4	Requires linking children's records to adult records 30–40 years later (Census Linking Project).
Table 5, Panel A (N discrepancy)	Table 5, Panel A	The released `FertilizersData.csv` lacks the 1925 agricultural census wave. Our $N = 5{,}912$ (two periods) vs. paper's $N = 8{,}983$ (three periods). All seven columns run but coefficients diverge in sign.
Figure 3: Event study	Figure 3	Requires the linked individual-level migrant panel (same as Table 3).
Table 8: Moral values	Table 8	MFQ survey data from YourMorals.org are restricted to the research team.

9.1.3 Outlier Sensitivity

To Be Completed

[Placeholder: Winsorize SHI at the 1st and 99th percentiles and re-run Table 1 Panel A and Panel D. Check whether results are driven by a small set of extreme-SHI counties (e.g., Appalachian or Great Plains outliers). Report Cook's D or influence statistics for the preferred specification.]

9.1.4 Alternative Standard Error Treatment

To Be Completed

[Placeholder: Compare results under (a) HC1 (heteroskedasticity-robust, no clustering); (b) spatial HAC clustering with a 200 km Conley radius; (c) two-way clustering by state and year. Discuss implications for statistical inference relative to the paper's grid100m clustering.]

9.2 Extension Analysis

To Be Completed

[Placeholder: Design and present at least one new empirical test that further supports or challenges the paper's main hypothesis. Example: Examine whether the negative SHI effect on LNI is stronger in counties with historically higher agricultural productivity (where social learning would have been more valuable), using GAEZ v3 suitability as a moderator. This would provide direct evidence for the social-learning mechanism as the channel, rather than other routes through which soil heterogeneity might affect community cohesion. Describe the test, present the results, and interpret them in the paper's framework.]

10 Research Design Critique

10.1 Do the Paper's Findings Support Its Claims?

To Be Completed

[Placeholder: Evaluate whether the empirical evidence in Raz (2025) convincingly supports the causal claim that soil heterogeneity shaped community formation through the social-learning mechanism. Discuss: (a) identification — is SHI plausibly exogenous to community cohesion, or could omitted variables (geographic isolation, access to markets, migration patterns) confound the relationship? (b) mechanism — does the Table 5 evidence convincingly isolate the social-learning channel, or could SHI affect cohesion through other routes (e.g., economic inequality from unequal land quality, differential selection of settlers)? (c) external validity — do the 19th-century US results apply to other contexts or time periods?]

10.2 What We Would Do Differently

To Be Completed

[Placeholder: Articulate specific improvements to the data choice, measurement, or empirical design. Examples: (a) use a spatial regression discontinuity design exploiting sharp geological boundaries (e.g., moraine edges) as natural experiments for soil heterogeneity variation; (b) use full-count census data (now available on FamilySearch for 1880–1940) rather than relying on IPUMS full-count agreements; (c) directly measure social interactions (e.g., intermarriage across soil-type boundaries within counties) rather than proxying community cohesion with naming patterns; (d) exploit variation in the timing of agricultural technology adoption to test whether counties with higher SHI adopted innovations more slowly after controlling for soil quality.]

10.3 Additional Analysis Under Alternative Design

To Be Completed

[Placeholder: If an alternative design is feasible with available data, implement it here and compare results to the paper's approach. Otherwise, articulate why the proposed design would be preferable and what data would be needed to implement it.]

11 Division of Work

To Be Completed

[Placeholder: Provide a brief paragraph describing the division of work among the four group members: Son Sang Hyuk, Chen Huaying, Chen Jiayi, and Guo Ruihua. Include who was responsible for data collection and preparation, variable construction (SHI, LNI, RHI, learning potential), regression analysis (Tables 1 and 5), figure production, and report writing.]

A Code Organization

Table 10 lists all R scripts in the `R/` directory, in the order they should be run. Each script is self-contained (sources `00_setup.R` at the start) and produces output to the `output/` directory.

References

Bergé, L. (2018). Efficient estimation of maximum likelihood models with multiple fixed-effects: The `R` package `fixest`. *CREA Discussion Paper*, 13:1–18.

Burbidge, J. B., Magee, L., and Robb, A. L. (1988). Alternative transformations to handle extreme values of the dependent variable. *Journal of the American Statistical Association*, 83(401):123–127.

FAO (2012). Global agro-ecological zones (GAEZ v3.0). Technical report, Food and Agriculture Organization of the United Nations and International Institute for Applied Systems Analysis, Rome, Italy.

FAO (2021). Global agro-ecological zones version 4 (GAEZ v4). Technical report, Food and Agriculture Organization of the United Nations, Rome, Italy.

Fryer, R. G. and Levitt, S. D. (2004). The causes and consequences of distinctively black names. *Quarterly Journal of Economics*, 119(3):767–805.

Manson, S., Schroeder, J., Van Riper, D., Kugler, T., and Ruggles, S. (2020). IPUMS National Historical Geographic Information System: Version 15.0. Minneapolis, MN: IPUMS. `http://doi.org/10.18128/D050.V15.0`.

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Ruggles, S., Flood, S., Foster, S., Goeken, R., Pacas, J., Schouweiler, M., and Sobek, M. (2020). IPUMS USA: Version 10.0. Minneapolis, MN: IPUMS. `https://doi.org/10.18128/D010.V10.0`.

Table 10: R Script Organization

Script	Purpose	Key output(s)
00_setup.R	Packages, paths, helpers (run_ols, load_panel)	Sourced by all scripts
01_build_shi_raster.R	Rasterize STATSGO2; 51×51 focal dissimilarity; county mean	SHI_v2.csv
02_build_geoclim_sd.R	Geo-climatic SD controls from HydroSHEDS + WorldClim	GeoClim_SD_county.csv
03_build_agri_suitability.R	GAEZ v3 suitability means/SDs for 10 crops	AgriSuitability_county.csv
04_build_rhi.R	NHGIS religion census; HHI; z-score within year	RHI_county_year.csv
05_build_learning_potential.R	gains5 and WheatDiff from GAEZ v4 + NHGIS 1930	LP_county.csv
06_data_description.R	Summary statistics	table_summary_stats.docx
07_figure1_shi_map.R	Choropleth map of SHI (state-FE residualized)	figure1_shi_map.png
08_table1_main_results.R	Table 1 Panels A and D (6 columns each)	table1_panel_a_lni.docx, table1_panel_d_rhi.docx
09_table5_mechanisms.R	Table 5 all 7 columns, both panels	table5_mechanisms.docx
10_robustness_extensions.R	Robustness and extension analysis	table_robustness.docx

Note: Scripts 01–05 are data-building scripts that require raw input files (STATSGO2 shapefile, GAEZ rasters, NHGIS CSVs) and take significant runtime (script 01: ≈40 min; scripts 02–05: <5 min each). Scripts 06–10 require only the processed CSV files in `data/` and can be run in minutes. All scripts use absolute paths defined in `00_setup.R`.